Learning Python

- `Counter(iterable)` (from `collections`) maps each element to its frequency and is the go-to tool for frequency counting in interview problems.
- `most_common(n)` returns the n most frequent `(element, count)` pairs in descending order (omit n for all), making it ideal for top-k frequency problems.
- `"".join([char for char in s if char.isalnum()])` strips a string to only letters and digits using `isalnum()`, which returns `True` for a–z, A–Z, and 0–9.
- `char.isnumeric()` accepts any Unicode numeric character (including fractions and superscripts), while `isdigit()` is stricter (ASCII digits 0–9 only).
- Never call `pop(i)` inside a loop that increments `i` forward, as each removal shifts remaining indices; a backward loop such as `for i in range(k, -1, -1): nums.pop(i)` is safe.
- `del nums[i]` is slightly faster than `nums.pop(i)` at the same O(n) complexity because `pop` has the extra overhead of returning the removed value, which can matter in tight loops over large lists.
- `list.index(i)` returns the index of the first occurrence of value `i` in the list, raising `ValueError` if it is not found.
- `list.count(value)` returns the number of times `value` appears in the list.
- `math.prod(list)` (from `math`) returns the product of all elements in the iterable, returning `1` for an empty iterable.
- `arr.insert(index, new_value)` inserts `new_value` at the given `index`, shifting all subsequent elements one position to the right.
- `my_dict.values()` returns a view object of all values in the dictionary, which can be saved into a list with `values_list = my_dict.values()`.
- `sum(my_dict.values())` returns the sum of all values in the dictionary, useful for quickly computing totals without an explicit loop.

- `max(my_dict.values())` returns the largest value in the dictionary, useful for finding the maximum frequency or count in a frequency map.
- `round(number, ndigits)` rounds `number` to `ndigits` decimal places (default 0), returning an `int` when called with no `ndigits` argument or a `float` otherwise.
- A **stack** in Python is simply a `list` where `append(val)` acts as push and `pop()` (no argument) acts as pop, both in O(1) time.
- Prefer `stack.append()` and `stack.pop()` over slice assignment such as `s = s[:-1]`, which creates a full O(n) copy each time and causes quadratic runtime in a loop.
- Two separate `if` statements in a loop are evaluated independently (both can trigger in the same iteration), whereas `if/elif` ensures only one branch fires per iteration.

1 LeetCode Practice

Table 1: LeetCode practice tracker

Question	Level	Revisit	Done
1. Arrays & Hashing			
217. Contains Duplicate	Easy		✓
1. Two Sum	Easy		✓
242. Valid Anagram	Easy		✓
383. Ransom Note	Easy		✓
349. Intersection of Two Arrays	Easy		✓
1512. Number of Good Pairs	Easy		✓
205. Isomorphic Strings	Easy		✓
49. Group Anagrams	Medium		✓
347. Top K Frequent Elements	Medium		✓
238. Product of Array Except Self	Medium		✓
2. Two Pointers			
344. Reverse String	Easy		✓
125. Valid Palindrome	Easy		✓
27. Remove Element	Easy		✓
26. Remove Duplicates from Sorted Array	Easy		✓
283. Move Zeroes	Easy		✓
977. Squares of a Sorted Array	Easy		✓
88. Merge Sorted Array	Easy		✓
167. Two Sum II: Input Array Is Sorted	Medium		✓
11. Container With Most Water	Medium		✓
15. 3Sum	Medium		✓
3. Sliding Window			

Question	Level	Revisit	Done
643. Maximum Average Subarray I	Easy		✓
1876. Substrings of Size Three with Distinct Characters	Easy		✓
2269. Find the K-Beauty of a Number	Easy		✓
1984. Minimum Difference Between Highest and Lowest of K Scores	Easy		✓
1652. Defuse the Bomb	Easy		✓
219. Contains Duplicate II	Easy		✓
121. Best Time to Buy and Sell Stock	Easy		✓
3. Longest Substring Without Repeating Characters	Medium		✓
567. Permutation in String	Medium		✓
424. Longest Repeating Character Replacement	Medium		✓
4. Stack			
20. Valid Parentheses	Easy	⋆	
682. Baseball Game	Easy		✓
1047. Remove All Adjacent Duplicates In String	Easy		✓
1544. Make The String Great	Easy		✓
844. Backspace String Compare	Easy		✓
232. Implement Queue using Stacks	Easy	⋆	
496. Next Greater Element I	Easy	⋆	
155. Min Stack	Medium		
150. Evaluate Reverse Polish Notation	Medium		
739. Daily Temperatures	Medium		
5. Binary Search			
704. Binary Search	Easy		
374. Guess Number Higher or Lower	Easy		
35. Search Insert Position	Easy		
278. First Bad Version	Easy		
744. Find Smallest Letter Greater Than Target	Easy		
367. Valid Perfect Square	Easy		
69. Sqrt(x)	Easy		
74. Search a 2D Matrix	Medium		
875. Koko Eating Bananas	Medium		
153. Find Minimum in Rotated Sorted Array	Medium		
6. Linked List			
876. Middle of the Linked List	Easy		
83. Remove Duplicates from Sorted List	Easy		
203. Remove Linked List Elements	Easy		
206. Reverse Linked List	Easy		
141. Linked List Cycle	Easy		
160. Intersection of Two Linked Lists	Easy		
234. Palindrome Linked List	Easy		
19. Remove Nth Node From End of List	Medium		
2. Add Two Numbers	Medium		
143. Reorder List	Medium		
7. Trees (DFS and BFS)			

Question	Level	Revisit	Done
94. Binary Tree Inorder Traversal	Easy		
104. Maximum Depth of Binary Tree	Easy		
226. Invert Binary Tree	Easy		
100. Same Tree	Easy		
101. Symmetric Tree	Easy		
112. Path Sum	Easy		
543. Diameter of Binary Tree	Easy		
102. Binary Tree Level Order Traversal	Medium		
98. Validate Binary Search Tree	Medium		
230. Kth Smallest Element in a BST	Medium		
8. Heap / Priority Queue			
1464. Maximum Product of Two Elements in an Array	Easy		
1046. Last Stone Weight	Easy		
2558. Take Gifts From the Richest Pile	Easy		
506. Relative Ranks	Easy		
1337. The K Weakest Rows in a Matrix	Easy		
2335. Minimum Amount of Time to Fill Cups	Easy		
703. Kth Largest Element in a Stream	Easy		
215. Kth Largest Element in an Array	Medium		
973. K Closest Points to Origin	Medium		
621. Task Scheduler	Medium		
9. Backtracking			
509. Fibonacci Number	Easy		
231. Power of Two	Easy		
326. Power of Three	Easy		
590. N-ary Tree Postorder Traversal	Easy		
257. Binary Tree Paths	Easy		
1863. Sum of All Subset XOR Totals	Easy		
401. Binary Watch	Easy		
78. Subsets	Medium		
46. Permutations	Medium		
39. Combination Sum	Medium		
10. Graphs			
1791. Find Center of Star Graph	Easy		
997. Find the Town Judge	Easy		
1436. Destination City	Easy		
733. Flood Fill	Easy		
463. Island Perimeter	Easy		
1971. Find if Path Exists in Graph	Easy		
993. Cousins in Binary Tree	Easy		
200. Number of Islands	Medium		
994. Rotting Oranges	Medium		
207. Course Schedule	Medium		
11. Dynamic Programming			
70. Climbing Stairs	Easy		

Question	Level	Revisit	Done
1137. N-th Tribonacci Number	Easy		
746. Min Cost Climbing Stairs	Easy		
118. Pascal's Triangle	Easy		
119. Pascal's Triangle II	Easy		
338. Counting Bits	Easy		
392. Is Subsequence	Easy		
198. House Robber	Medium		
322. Coin Change	Medium		
300. Longest Increasing Subsequence	Medium		